

ENGR 3430: Miniproject 3

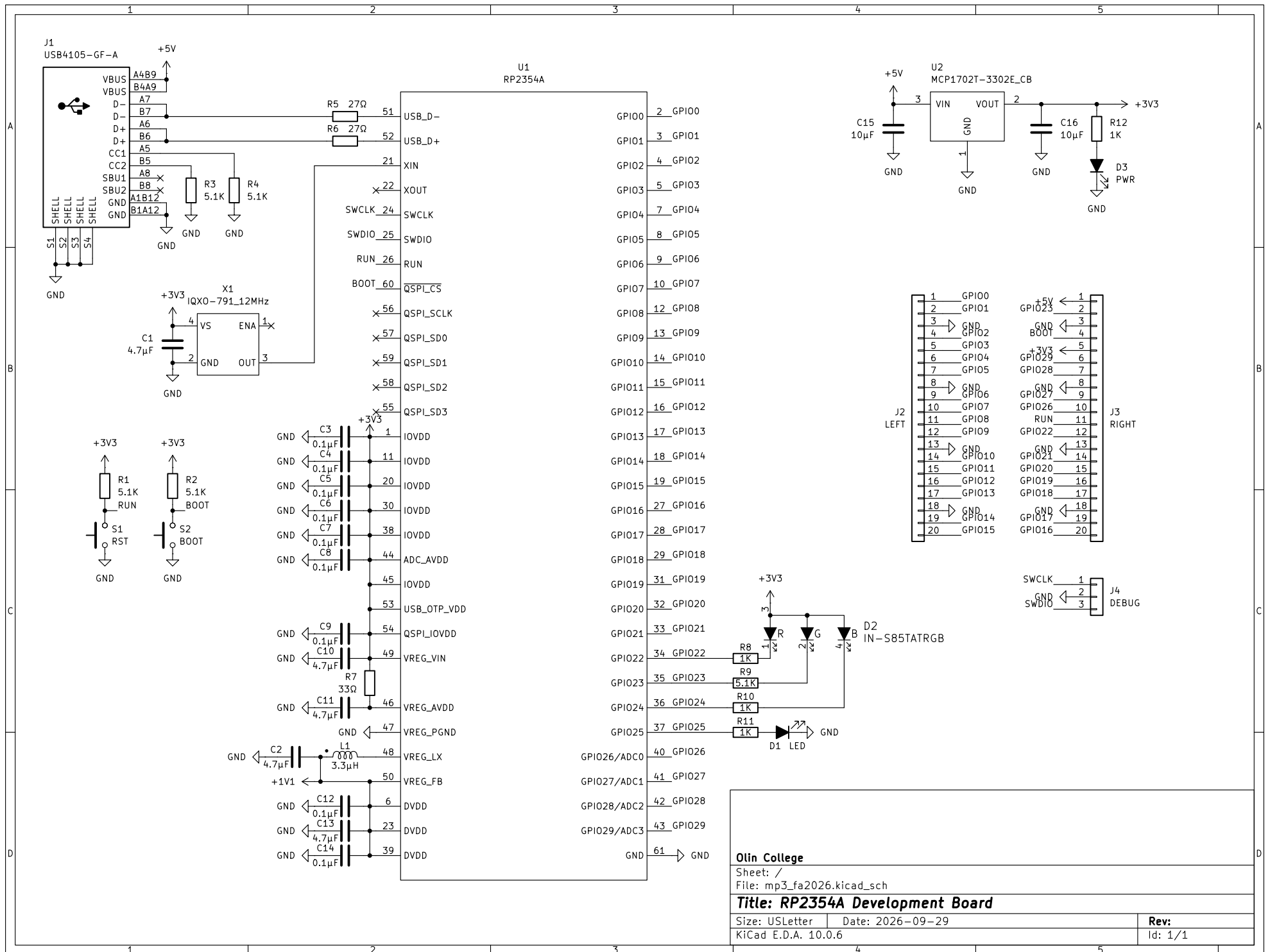
layout due October 16, 2026
demo due November 13, 2026

In this miniproject, you will gain more experience doing surface-mount printed circuit board (PCB) layout and assembly with a more complex circuit. You will be designing a Raspberry-Pi-Pico-2-like microcontroller development board, which is shown in the attached schematic, based on a KiCad project that you can download from the “Files” section of the course Canvas site. Other than a maximum size constraint of about 4 in $\times$ 4 in imposed by the board fabricator (PCBWay), there are no externally imposed board size considerations. For the tactile switches, resistors, and most capacitors, we will have two size options available for you use to assemble your boards. The list of available components for MP3 will be provided as a spreadsheet that you can download from the course Canvas site. We will submit your boards for fabrication through PCBWay. When they come back, you will assemble your board, program the microcontroller to blink the user LED at 1 Hz and submit a demo video of the code running on your board.

Requirements. Your design must meet the following requirements:

1. It must be a two-layer board, no larger than 3.94 in $\times$ 3.94 in (i.e., 100 mm $\times$ 100 mm) with all components on the top side.
2. The minimum allowable trace width and spacing is 6 mils.
3. The minimum allowable via size is 24 mils with a 12-mil drill hole.

Deliverables. Your final PCB layout is due by the start of class on Friday, October 16. You will need to submit copies of all of your KiCad files. Assuming there are no problems with your design, we will generate the Gerber files and submit them to PCBWay for fabrication. The boards should be back from fab by the end of October. Once they return, you will assemble your board, get an appropriate firmware development toolchain up and running to write code, and demonstrate the blinking LED running on your board by no later than the start of class on Friday, November 13.



Olin College

Sheet: /
File: mp3_fa2026.kicad_sch

Title: RP2354A Development Board

Size: USLetter Date: 2026-09-29
KiCad E.D.A. 10.0.6

Rev:
Id: 1/1

Item	Reference	Description	Manufacturer's Part #	Digikey Catalog #	Qty
1	C3, C4, C5, C6, C7, C8, C9, C12, C14	0402, 0.1-μF, 50-V, ±5%, X5R ceramic capacitor	0402X104K500CT	1292-1366-1-ND	9
		0603, 0.1-μF, 50-V, ±5%, X7R ceramic capacitor	CL10B104JB8NNNC	1276-1033-1-ND	
2	C1, C2, C10, C11, C13	0402, 4.7-μF, 10-V, ±10%, X5R ceramic capacitor	CL05A475KP5NRNC	1276-1480-1-ND	5
		0603, 4.7-μF, 16-V, ±10%, X7R ceramic capacitor	GRM188Z71C475KE21D	490-GRM188Z71C475KE21DCT-ND	
3	C15, C16	0603, 10-μF, 25V, ±10%, X5R ceramic capacitor	GRM188R61E106KA73D	490-18214-1-ND	2
4	R1, R2, R3, R4, R9	0402, 5.1-kΩ, 1/16-W, ±1% resistor	AC0402FR-075K1L	YAG3503CT-ND	5
		0603, 5.1-kΩ, 1/10-W, ±1% resistor	ERJ-3EKF5101V	P5.10KHCT-ND	
5	R8, R10, R11, R12	0402, 1-kΩ, 1/16-W, ±1% resistor	CRGCQ0402F1K0	A129627CT-ND	4
		0603, 1-kΩ, 1/10-W, ±1% resistor	RC0603FR-071KL	311-1.00KHRCT-ND	
6	R5, R6	0402, 27-Ω, 1/16-W, ±1% resistor	ERJ-2RKF27R0X	P27.0LCT-ND	2
		0603, 27-Ω, 1/10-W, ±1% resistor	RC0603FR-0727RL	311-27.0HRCT-ND	
7	R7	0402, 33-Ω, 1/16-W, ±1% resistor	RC0402FR-0733RL	311-33.0LRCT-ND	1
		0603, 33-Ω, 1/10-W, ±1% resistor	RC0603FR-0733RL	13-RC0603FR-0733RLCT-ND	
8	L1	3.3-μH, 2.1-A, 140-mΩ inductor	AOTA-B201610S3R3-101-T	535-AOTA-B201610S3R3-101-TCT-ND	1
9	D1	0805 user LED (orange)	XZCM2MOK54WA-1VF	1497-XZCM2MOK54WA-1VFCT-ND	1
10	D2	0805 RGB LED (common anode)	IN-S85TATRGB	1830-1086-1-ND	1
11	D3	0805 power indicator LED (green)	LTST-C170TGKT	160-1887-1-ND	1
12	U1	RP2354A microcontroller	SC1511(13)-A4	2648-SC1511(13)-A4CT-ND	1
13	U2	3.3-V LDO linear regulator	MCP1702T-3302E/CB	MCP1702T-3302E/CBCT-ND	1
14	X1	12-MHz CMOS oscillator	831056289	1923-831056289CT-ND	1
15	S1, S2	Tactile switch	B3U-1000P	SW1020CT-ND	2
		Tactile switch, SPST-NO, 0.05-A, 12-V	TL3301AF160QJ	EG4374CT-ND	
16	J1	USB 2.0 type C connector	USB4105-GF-A	2073-USB4105-GF-ACT-ND	1